COLORADO · THE NEXT ERA OF COMPUTE

Silicon Valley > Photon Mountain

Photon Mountain

Silicon Valley was named for the material of the last era. Photon Mountain is named for the medium of the next, light. Along Colorado's Front Range, one corridor is building the AI age out of photons: light-based chips, renewable energy, and quantum systems.

WHAT IS PHOTON MOUNTAIN

Photon Mountain is a real place, the Front Range corridor where Colorado builds the AI age out of light. A photon is the particle of light, so the name holds both the photonic chips Colorado can make and the quantum science it can run. Two rings: a dense core of labs and fabs, and a greater ring of cities.

CORE The dense Boulder, Golden, Arvada, and Denver cluster: the labs, the fabs, and the metro hub.



GREATER The Front Range ring around the core, where the corridor's companies and workforce spread out.

Lafayette · Louisville · Broomfield · Westminster · Longmont

PHOTONIC CHIPS

Chips that compute and move data with light, at far lower power. Colorado fabricates them today at NIST Boulder.

>$13B silicon photonics market by 2034

RENEWABLE ENERGY

Power is the constraint on the AI build. Golden's National Laboratory of the Rockies (formerly NREL) is the nation's renewable-energy lab.

100% renewable-electricity goal by 2040 (Polis roadmap)

QUANTUM SYSTEMS

The Elevate Quantum hub, and the nation's first quantum-engineering degree at Colorado School of Mines.

$127M+ federal and state investment to date

SILICON VALLEY	PHOTON MOUNTAIN
The computer age	**The AI age**
Silicon, the material	**Photon, the light**
San Jose to Menlo Park	**Denver to Boulder**
Microchips	**Photonic chips, renewable energy, quantum**

Silicon Valley built the last era out of silicon. Photon Mountain builds the next out of light.

Build with light.

Photon Mountain · a Colorado Startups project · paul@coloradostartups.org

Sources: NIST; Elevate Quantum / Colorado OEDIT; Colorado School of Mines; National Laboratory of the Rockies (formerly NREL); Colorado Energy Office and Xcel Clean Energy Plan. NIST Boulder is an operating photonic-chip cleanroom; the Quantum Commons (Arvada) and CU Boulder fabs open in 2026. Colorado's 100% renewable-electricity target is a 2040 goal. 2025-2026.